Optimizing Operational Efficiency and Seasonal Profitability: A Data-Driven Strategy for an Air Conditioner Repair Center

A Final report for the BDM capstone Project

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1. Executive Summary –

This report presents a detailed data driven study on **Aone Cooling Center**, a mid-sized air conditioner servicing business, aimed at resolving two critical operational problems that are **procurement cost management** and **hired workforce management**. The project uses authentic data provided by the business owner, covering operations from **January 2023 to December 2024**, and applies statistical modeling, forecasting, and optimization techniques to derive actionable strategies.

The starting phase involved digitizing handwritten records and seller invoices related to procurement and workforce scheduling. The raw data, once structured in spreadsheets, was cleaned and processed using Python-based tools on Google Colab. Libraries such as Pandas and NumPy were used to standardize formats, handle missing values, and validate entries, ensuring the reliability of subsequent analysis.

For **procurement management**, seasonal trends were examined by dividing each year into three segments: **January–March**, **April–July**, and **August–December**. Descriptive statistics and visualizations uncovered predictable spikes in demand during the **April–July** period. This spike coincided with noticeable increases in unit costs particularly for essential parts such as **R-22** and **Hydrocarbon**, which rose by up to **18%** during the peak season.

To tackle the above problem and further strengthen inventory planning, the **ARIMA time series forecasting model** was implemented. Beginning with a focused case study on **R-22**, the model was later applied to the broader inventory dataset. It forecasted demand for the first two quarters of 2025, offering reliable insights that help align procurement with expected usage and seasonal pricing. This helped prevent overstocking during low demand periods and reduced emergency procurement costs during seasonal peaks.

In terms of **workforce optimization**, the repair center had been operational with a fixed team of 8 workers, irrespective of seasonal fluctuations in service requests. This led to both **understaffing during peak months** and **idle worker costs during off-peak months**. Using the monthly service request data, the required number of workers is calculated.

To find a cost-effective staffing structure, a **Linear Programming (LP) model using the Simplex Method** was implemented. The model minimized worker costs while ensuring service demand was met each month. However, recognizing the limitations of a purely mathematical approach, a **hybrid staffing strategy** was proposed: maintaining **7 permanent workers** year-round and hiring additional staff on a **contractual basis** during peak seasons. This approach ensures workforce flexibility while minimizing operational costs.

In conclusion, the study transforms operational data into a highly efficient strategy framework. By combining **trend analysis**, **predictive modeling**, and **mathematical optimization**, it delivers practical recommendations that allows Aone Cooling Center to align spare part procurement and workforce planning with real-world demand cycles. The outcomes support **cost reduction**, **enhanced efficiency**, and **ineffectiveness towards seasonal fluctuations**, making the business free and more competitive in an unstable service environment.

2. Detailed Explanation of Analysis –

The detailed analysis of data for this project was conducted systematically, in order to resolve the core issues identified in the project namely procurement cost inefficiencies and irregular workforce hiring, a structured data analytical process was followed. The aim was to extract actionable insights from historical records that could support smarter procurement planning and more efficient staffing, more specifically in response to seasonal variations in demand. The overall process included data cleaning, preprocessing, exploration, modeling, and strategy formulation.

Data cleaning and Preprocessing

The raw procurement and workforce data were collected from historical usage of spare part records maintained on paper, seller bills, and some are taken by the owner's description. The data was structured in spreadsheets and then imported into Google Colab for further analysis using Python libraries such as pandas and NumPy.

For ensuring analytical accuracy, data integrity checks were performed widely ensuring all data types (e.g., integers for costs and dates for service records) were consistent and correctly formatted, including replacing missing values, and validating demands, cost fluctuation workforce(hired) records.

This preprocessing ensured us that the dataset is now reliable and ready for deeper statistical and predictive analysis.

Descriptive Statistics

The subsequent phase involved conducting descriptive statistics on the data. Descriptive statistics were applied to gain insights into the distribution and variation within the dataset. The main statistical measures were calculated which includes various measures such as mean, standard deviation, minimum, and maximum values etc., which in turn to the better understanding with the trends of the market and distribution and variation of the fields in the dataset. This was facilitated by the describe () and info () methods of the Pandas dataframe. These insights provided me with a preliminary understanding of trends and variability, such as which months experienced higher demand, and the spread in procurement costs over time.

The analysis involved in addressing each problem statement is as follows

Tackling Problem 1: Managing Procurement Costs and Streamlining Inventory

To manage procurement costs effectively and optimize inventory levels, the first step involved ordering and filing historical data on spare parts demand including quantities used, dates, price variation etc., and prices of different spare parts in different parts of the previous year based on different seasons. This data was collected from existing hand written records about usage of spare parts in each service done and the seller bills of buying in different seasons. After compiling the data in spreadsheets, the data is imported into Google Colab, and then data preprocessing was carried out using Python. This involved converting text dates to proper datetime formats, standardizing missing values, and verifying numerical continuity in procurement costs and quantity records.

Descriptive statistics were used in first place to explore the range, mean, and seasonal variation in demand. This helped me understand which months experienced the highest service requests and which spare parts were used most frequently. I then segmented the year into three separate parts: January–March (off-season), April–July (peak summer season), and August–December (post-monsoon to winter) which is done with the reference of seasons.

Then on, we started by comparing the **seasonal demand trends across 2023 and 2024**. The idea behind this was to highlight patterns such as consistently high demand in specific months or sharp deviations between the years. By visualizing like line graphs, bar plots and tabulating these trends, we could predict likely demand for upcoming months and estimate how much stock would be required ahead of time.

Next, we analyzed historical procurement prices to detect fluctuations and identify optimal periods for cost-effective purchases. A line graph of monthly prices was generated to visualize low-cost windows, allowing the business to strategically schedule bulk purchases when prices were lowest. We also calculated reorder points for each part, based on average consumption and lead time, to ensure timely restocking without excess inventory. This two-fold analysis not only reduced unnecessary procurement costs but also helped prevent stockouts during peak demand.

To enhance forecasting accuracy, a time series model was implemented using ARIMA. This model, trained on the historical monthly demand data, was used to predict future needs. The steps included:

Differencing the series to make it stationary.

Determining optimal p, d, q values using ACF/PACF plots.

Fitting the ARIMA model and conducting **residual analysis** to check model performance.

The forecasts allowed the business to plan procurement cycles in advance, matching stock levels with actual demand patterns across different months. The ARIMA model's predictions supported long-term planning and ensured purchasing decisions were data-driven rather than guesswork-based.

These predictions supported smarter purchasing decisions, helping avoid overstocking in off-seasons and shortages during high-demand months.

Tackling Problem 2: Optimizing Hired Workforce Based on Demand

The next major challenge I faced was about how much Workforce need to be hired based on seasonal demand to avoid both understaffing and overstaffing resulting in unnecessary labor costs. The main focus was on developing a workforce management strategy that could adapt to seasonal service fluctuations while maintaining high service quality (which is basically calculating minimum workforce to be hired because we can't hire them just for a month).

Historical records of service requests were examined month by month, Monthly service demand showed a clear peak during the summer season, tapering off in cooler months. Along with this, data on worker schedules, salaries, and service capacity (i.e., number of services one worker can handle in a month) were included. The available data on service requests in a month, worker capacity of workers, and salary per worker was used in order to design a staffing plan that responds to seasonal service loads.

For each month, we calculated the required number of workers using the formula:

Required Workers = Total Service Requests / Requests Handled Per Worker

This value was compared to the actual number of hired workers to determine the:
Worker Shortage (When required workers > hired workers) and Worker Excess (When hired workers > required workers) using formula:

Worker Shortage/Excess = | Hired Workers – Required Workers |

Using this data, a Linear Programming model was developed with the target of minimizing total labor cost while ensuring that the demand was met adequately each month. The model took into account constraints such as the maximum number of services a technician could handle per month, the minimum workforce required during peak periods and the cost per worker.
By using the Simplex Method, we determined the optimal number of workers to hire for each month. The results suggested staffing in summer while reducing it in off-peak periods, which helped cut down idle labor costs and service delays. This data-driven approach allowed us to plan staff numbers dynamically, reducing idle wages in slower months while ensuring sufficient manpower during busy periods. This approach also ensured that service delivery remained uninterrupted during high demand periods, improving customer satisfaction without overspending on human resources. The result was a leaner workforce strategy that matched operational needs more closely, maintaining service quality and reducing unnecessary expenses.

Using this trend, we modeled labor needs in a way that ensures neither over-hiring nor staff shortages. Instead of fixed staffing models that fail to adapt to changing needs, this strategy enabled dynamic workforce planning that aligned with real business conditions.

By conducting targeted analysis for both procurement and workforce planning, we developed a more adaptive and cost-effective operational model for the air conditioner service center. Procurement decisions are now better aligned with actual usage and pricing trends, while staffing levels are dynamically planned according to seasonal workload. The strategies were built on real operational data, using statistical and mathematical techniques that ensure the recommendations are realistic and beneficial for the business in the long term.

3. Results and Findings –

This section presents the outcomes of the data analysis, predictive modeling, and optimization techniques implemented to address the two major operational problems faced by Aone Cooling Center i.e., procurement cost inefficiencies and seasonal workforce imbalances. The analysis of procurement costs factors and workforce hiring requirements and usage, revealed several key trends that impact operational efficiency and profitability. Insights were derived from trends analysed in spreadsheets and descriptive statistics, time series forecasting (ARIMA), and linear programming optimization, all conducted in Google Colab using Pandas and NumPy libraries of python.

(Here is my Google colab link for reference: Colab File)

Following is a detailed breakdown of findings, supported by graphs and interpretations –

Problem Statement 1: Procurement Cost Management and Inventory Optimization

The first step in understanding procurement patterns involves analyzing the seasonal demand variations between the years 2023 and 2024.

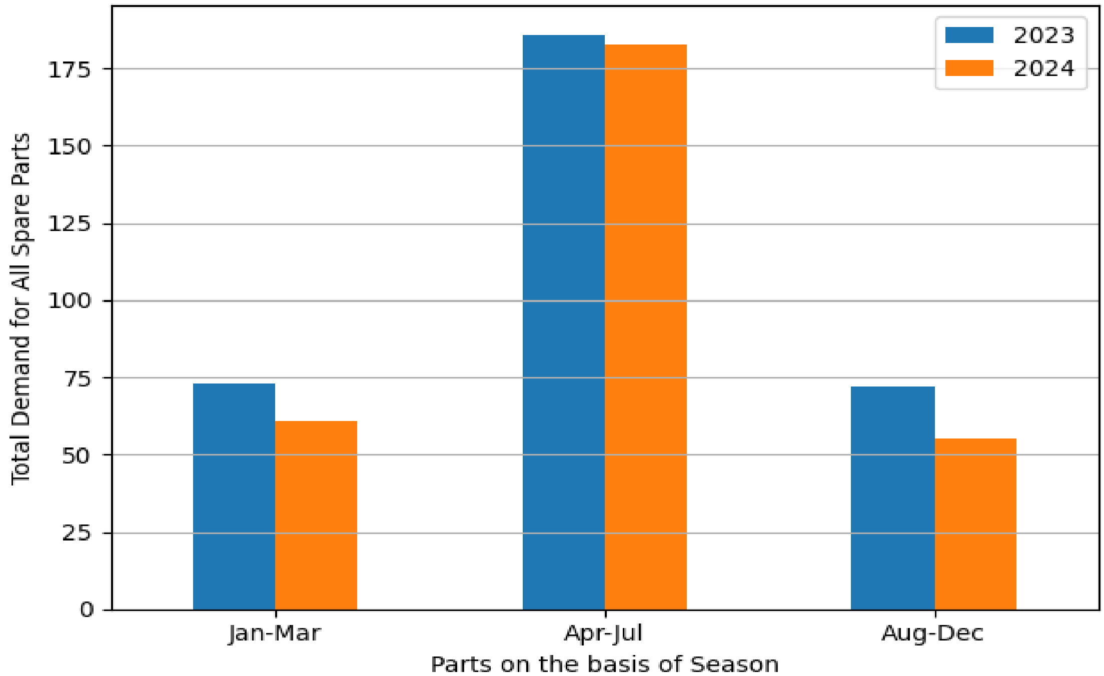
Seasonal Demand Analysis

The primary aim was to identify consistent high-demand periods and detect any unusual shifts between the years. I did this on colab, first of all I sum up the demand of all spare parts in the three parts I broke the year in, then I created a bar graph comparing the demands from 2023 and 2024 as well as showcasing the portion of the year with the high demand of spare parts among all the parts.

Figure 1 displays a bar chart summarizing the combined demand for all spare parts across three key seasonal segments: January–March, April–July, and August–December.

In 2023, the highest total demand was recorded during **April–July** with **186 units**, followed by **January–March** at **73 units**, and **August–December** at **72 units**. A similar trend is observed in 2024, where **April–July** again shows the highest demand with **183 units**, while **January–March** and **August–December** recorded **61** and **55 units**, respectively. In contrast, the August–December period showed a slight reversal, where 2023 demand (73 units) surpassed that of 2024 (61 units).

Figure 1: Seasonal Demand Comparison: 2023 vs 2024



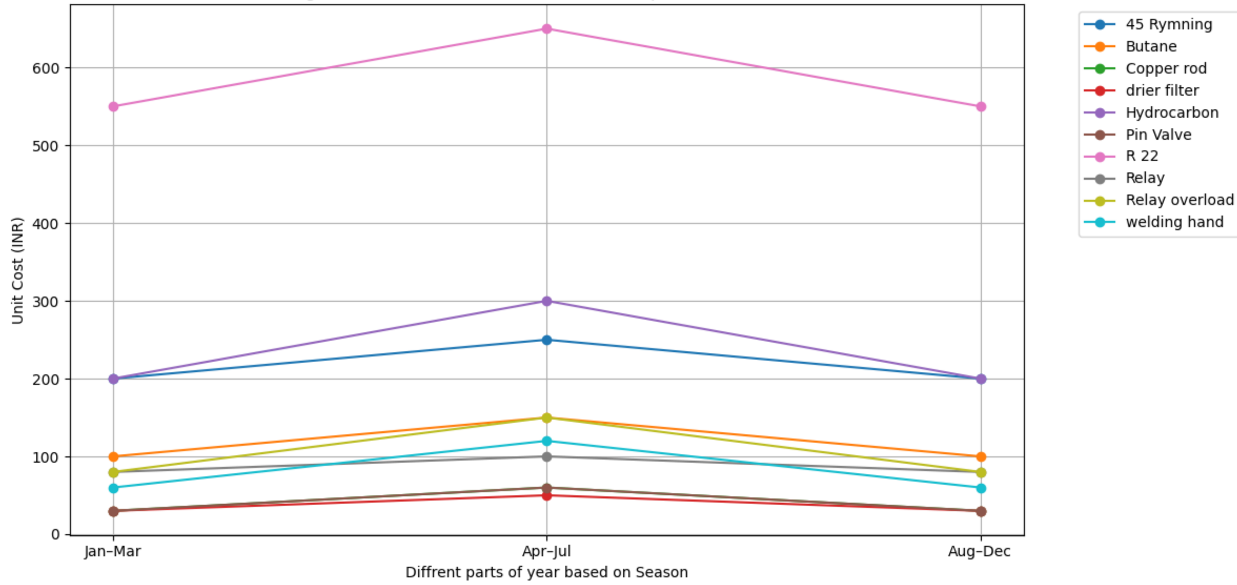
Although the overall demand in 2024 was slightly less across all three seasons, the pattern of peak consumption during **April–July** remains consistent. This recurring seasonal spike highlights the importance of planning procurement strategies well in advance particularly during the final months of the preceding year when prices are potentially more favorable.

This analysis plays a very important role in making inventory decisions. By understanding behavior seasonally demand, the business can stock essential components before time, minimizing both shortages and cost surges during high-demand windows.

Unit cost Trends Analysis

Now we know about the months that have a high demand and then in order to better understand procurement cost behavior across the year, a seasonal analysis of unit prices was conducted for all key spare parts.

Figure 2: Seasonal Unit Price Trends for Spare Parts (2024)



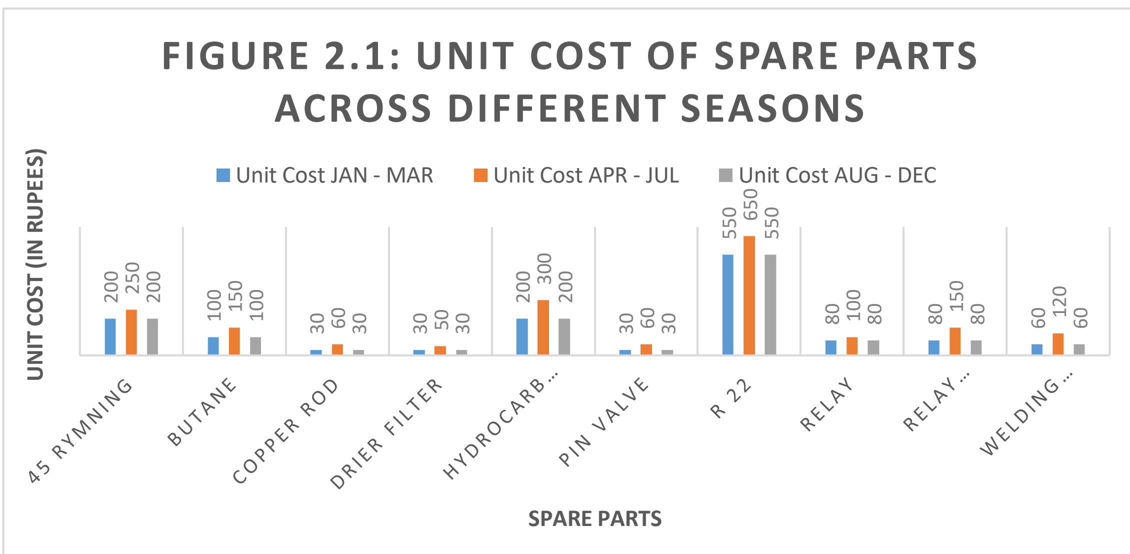


Figure 2 and 2.1 presents a line chart and a bar chart comparing unit costs for each item across the three major seasonal periods of 2024: **January–March**, **April–July**, and **August–December**.

The charts reveal a noticeable increase in prices during the **April–July** period for almost all items. For instance, the cost of R-22 rose from ₹550 in **Jan–Mar** to ₹650 in **Apr–Jul**, before returning to ₹550 in **Aug–Dec**. A similar pattern was observed in other parts such as Hydrocarbon and Pin Valve, both of which showed a significant rise in the second quarter. In contrast, parts like copper rod, Drier filter, and Relay overload maintained relatively stable prices throughout the year.

This mid-year spike in prices strongly suggests that **April–July is the least favorable time for bulk purchasing**, likely due to seasonal demand pressures or vendor pricing behavior. Based on this trend, it is recommended that the repair center **schedule major procurements before the end of March**, particularly for high-demand or high-cost items. Doing so can help the business lower its overall procurement costs and avoid financial strain during the busier service periods.

By aligning purchasing schedules with cost-efficient periods, the business not only reduces expenses but also ensures the timely availability of critical components. This cost-aware approach to inventory procurement supports long-term financial sustainability while minimizing the risks of stock shortages or overpriced emergency buys.

Forecasting Spare Part Demand Using ARIMA

To bring more accuracy and clarity into inventory planning, a time series forecasting model was implemented using the ARIMA (AutoRegressive Integrated Moving Average) technique. I developed ARIMA model that predict quarterly demand for spare parts and was trained on historical data of spare parts that showed seasonal trends from 2023 and 2024.

Model Overview

The ARIMA model was configured as ARIMA (1,1,1) –

AR (p=1): Captures the influence of previous demand values.

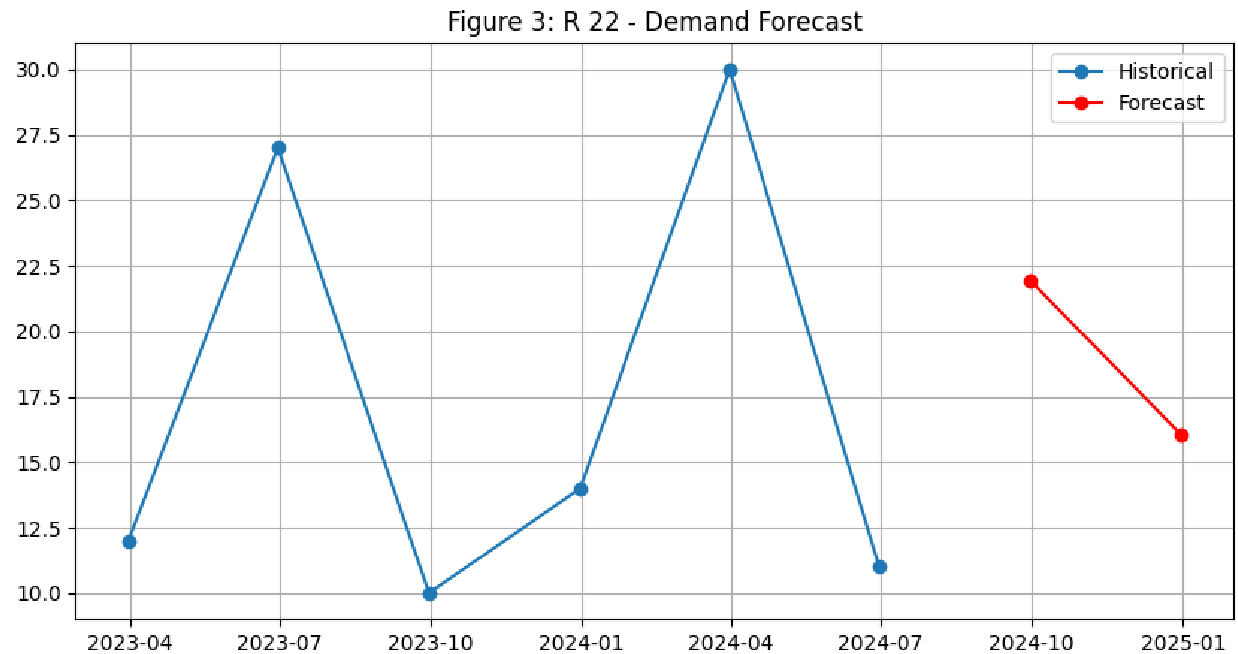
I (d=1): Applies first-order differencing to remove trend and seasonal effects, ensuring stationarity.

MA (q=1): Adjusts for past errors to refine the forecast.

I first illustrate this approach with a detailed case study of R-22 and then extend the analysis to all spare parts.

Detailed Forecast for R-22

R-22, being a commonly used refrigerant, was forecasted separately to illustrate how seasonal demand patterns affect procurement planning.



Forecasted Demand:
2024-09-30 21.948227
2024-12-31 16.028735

The output model indicated that R-22 follows a moderate seasonal trend, with the forecast predicting a demand of approximately **21.95 units in Q1 2025** and **16.03 units in Q2 2025**. The historical and forecasted demand pattern for R-22 depicted in **Figure 3**, clearly showcase that The ARIMA model highlighted that demand typically peaks between **April and July**, underscoring the importance of **early procurement**. Since unit costs for R-22 increase from ₹550 in Jan–Mar to ₹650 in Apr–Jul, this forecast empowers the center to **order bulk quantities during the first quarter**, before prices rise.

Forecast Summary Across Spare Parts

While the R-22 forecast provides an insightful deep dive, it is equally important to consider the broader picture. We extended the ARIMA forecasting approach to all spare parts. For each component, a similar six-point quarterly demand series was used to fit an ARIMA (1,1,1) model, and forecasts were generated for Q1 and Q2 of 2025. The resulting forecasts are summarized in the table below.

index	Spare Part	Forecast_Q1_2025	Forecast_Q2_2025
0	45 Rymning	-6.15E-05	-6.15E-05
1	Butane	7.99999467	7.99999467
2	Copper rod	22.33765138	12.52781246
3	drier filter	9.999996478	9.999996478
4	Hydrocarbon	19.90797539	12.4322153

5	Pin Valve	8.999986329	8.999986329
6	R 22	21.94822725	16.02873536
7	Relay	7.954337176	7.66069031
8	Relay overload	9.028403767	4.143771266

Firstly, I removed welding hand because its value is 0.
This aggregated view highlights that, while several components (e.g., Copper rod, Hydrocarbon) most probably are expected to experience high demand in the upcoming quarter, others (such as 45 Rymning and Welding hand) show minimal forecasted usage. The differentiated forecasts underscore that procurement strategies should be tailored on a per-item basis rather than adopting a one fits all approach.

Managerial Implications

The two layered forecasting approach detailing both a single critical component and the overall product portfolio provide us a support for data driven procurement decisions:

Targeted Ordering: The detailed R-22 model indicates a slight forecast decline after a Q2 peak, which suggests that bulk ordering in Q1, when prices are favorable, can effectively secure inventory without overstocking.

Portfolio Optimization: The aggregate forecasts reveal varied demand profiles among spare parts. High-demand items like copper rod and Hydrocarbon require preemptive procurement, whereas items forecasted at near-zero demand can be deprioritized.

Cost Efficiency: By aligning purchasing strategies with these forecasts, the repair center can avoid emergency orders during high-cost periods and significantly reduce carrying costs.

Strategic Budgeting: The insight-driven forecasts empower precise budgeting and operational planning, ensuring a balance between inventory availability and financial prudence.

In essence, the integration of both a detailed forecast for a representative spare part and an aggregate forecast across the product range transforms procurement from a reactive to a predictive function—supporting improved inventory control, cost savings, and enhanced service reliability.

Problem Statement 2: Optimizing Hired Workforce Based on Demand

Another critical objective of the project was to design a staffing plan that adjusts to fluctuating service demands across the year, ensuring service continuity without incurring excessive labor costs. This was particularly important because overstaffing in low-demand months leads to idle wage payments, while understaffing during peak seasons results in delays and poor service quality.

Historical Workforce Demand Analysis

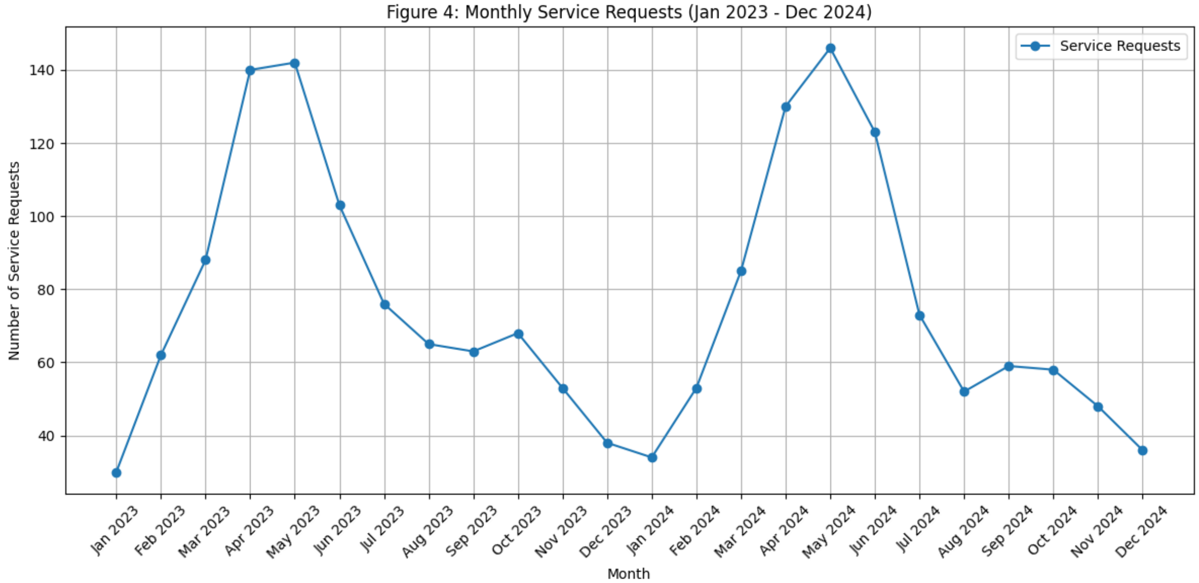
In order to understand the actual service load and staffing requirements, First of all I examined the data month wise spanning January 2023 to December 2024.The number of service requests for each month was divided by the fixed service capacity of 10 requests per worker to derive the required workers.

$$\text{Required Workers} = \frac{\text{Total Service Requests}}{\text{Requests Handled Per Worker}}$$

For example, if 120 service requests in January 2023, the calculated need was 12 workers (calculated as 120/10). This computation was carried out, revealing seasonal trends: the summer period (e.g., April and May 2023) showed higher demand with up to 14 workers needed, whereas low-demand months like December 2023 required as few as 4 workers. For a quick view here are first five entries of the transformed data which is edited for analysis followed by the Figure 4 showing Monthly Service Requests from Jan 2023 to Dec 2024 –

S.No.	Month	Servise Requests	Hired Workers	Request_handled_per_worker	Required Workers	Workers_Shortage	Workers _excess
1	Jan-23	30	8	10	3	0	5
2	Feb-23	62	8	10	6	0	2
3	Mar-23	88	8	10	9	1	0
4	Apr-23	140	8	10	14	6	0
5	May-23	142	8	10	14	6	0

Figure 4: Monthly Service Requests (Jan 2023 - Dec 2024)



This preliminary calculation highlighted that the existing fixed staffing level (8 workers per month) was not aligned with the actual demand, **Peak demand** was observed during the summer months (e.g., April and May 2023 showed requirements of 14 workers), indicating periods where service load intensifies. Conversely, **lower demand** periods (such as December 2023 with only 4 required workers) suggested that the existing fixed staffing level (8 workers per month) was excessive. Which in turn leads to both overstaffing during slow periods and under-staffing during peak periods.

Staffing Requirement and Cost Analysis

Next, we compared the actual hired worker that is 8 workers with the calculated required workers for each month.

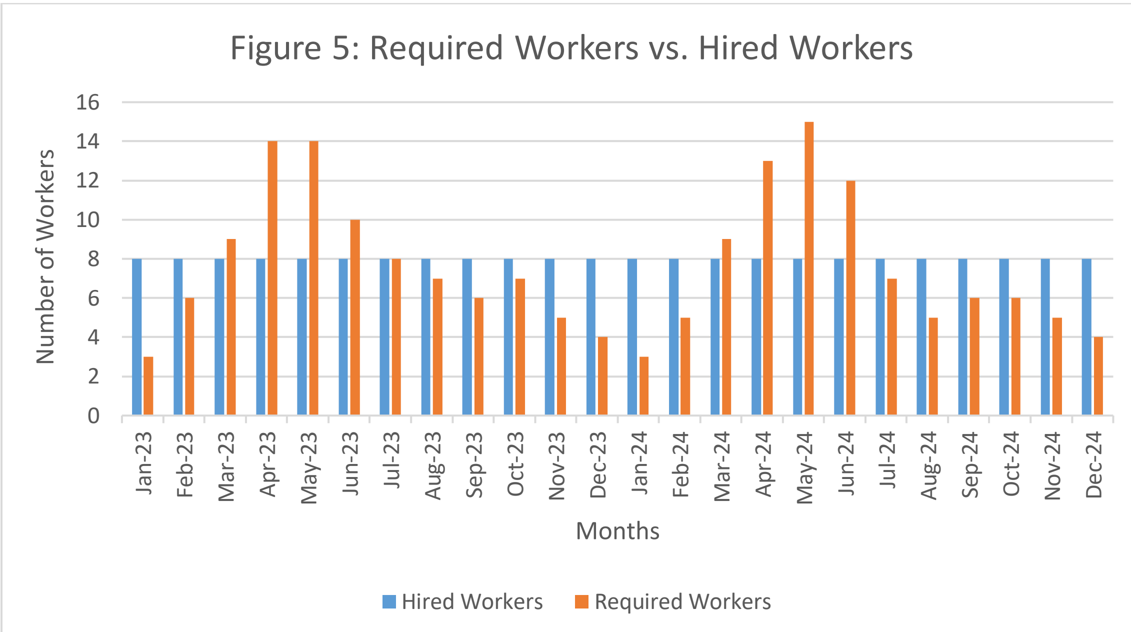


Figure 5 is a side-by-side bar chart visualizes both the "Hired Workers" and the "Required Workers" for each month. By doing so, we could identify staffing mismatches:

Worker Shortage: It occurs when the required number of workers exceeds 8 (e.g., 14 workers needed in April 2023).

Worker Excess: It occurs when the actual hired workforce is more than needed (e.g., 8 workers when only 4 are required in December 2023).

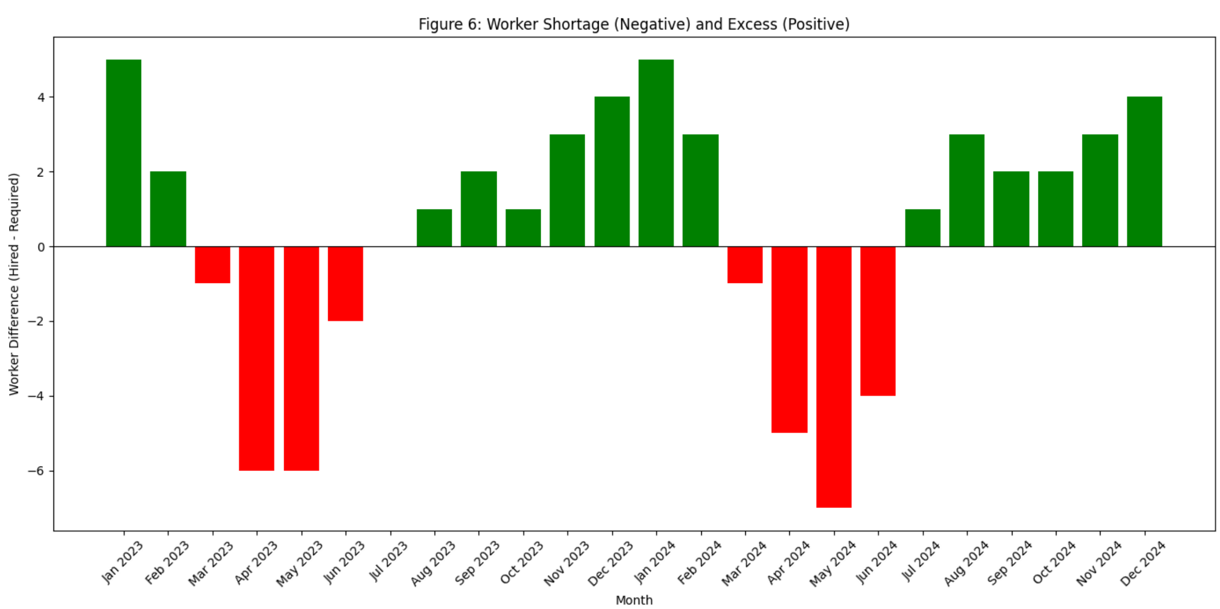


Figure 6 represents a bar or area chart showing the difference between the "Hired Workers" and "Required Workers" i.e., worker shortage or worker excess reinforces the cost implications of a static staffing model. This graph quantifies how many extra workers are paid during low demand or how many additional workers are needed during peak periods.

Given that worker expenses are ₹20,000 per worker per month, maintaining a constant high level of number of workers in periods of low demand results in unnecessary costs, while insufficient workers during peak periods could lead to service delays.

LP Model Fitting and Workforce Optimization

To address the misalignment between staffing and demand, a Linear Programming (LP) model was formulated and solved using the Simplex Method. The model's aim was to minimize overall labor costs while ensuring that the number of workers hired each month met or exceeded the calculated required levels. The LP formulation was as follows:

Objective Function:

$$\min \sum_{i=1}^{n} (\text{Cost per Worker} \times x_i)$$

where x_i is the optimized number of workers in month i, with a cost assumed at ₹10,000 per worker.

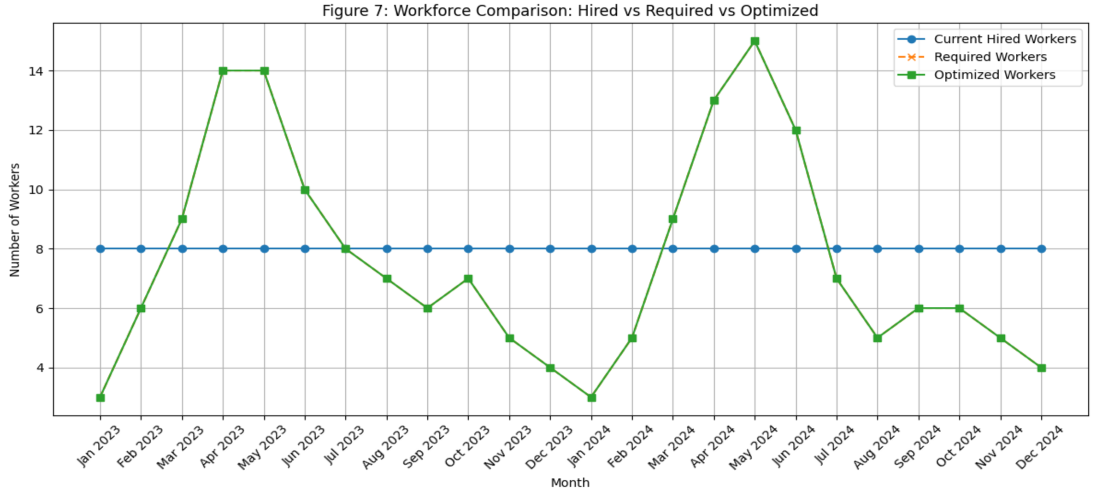
Constraints:

For every month i:

$$x_i \geq \text{Required Workers}_i$$

Figure 7 represents a plot that is generated to compare the three key metrics:
the constant number of hired workers, the calculated required workers, and the model's optimized workers. This visualization clearly shows the discrepancies in the current static model and how the LP approach results in a leaner, cost-effective staffing strategy.

Figure 7: Workforce Comparison: Hired vs Required vs Optimized



This calculation across the data revealed significant seasonal variations. For instance, demand required as few as 3 workers in low-demand months (e.g., January 2023) and as many as 14 during peak periods (e.g., April 2023).

The median of the required workers of all 24 months was approximately 7. This median value reflects the baseline staffing level needed to reliably serve routine service demand throughout the year.

A Hybrid Staffing Strategy

The most appropriate solution is to maintain a **permanent core team of 7 high skilled workers**. This team will be sufficient to cover the baseline level of service requests during low-demand periods, ensuring consistent service quality and avoiding unnecessary salaries.

During months when the required number exceeds 7, additional workers should be employed on a **contract basis**. For example:

In **April 2023**, the data indicates that 14 workers are needed. With a permanent staff of 7, an additional **7 contract workers** would fill the gap.

In **December 2023**, only 4 workers are required; hence, the permanent team covers this demand, and no contract hires are needed.

Figure 8 : Hybrid Workforce Model: Permanent and Contract Workers

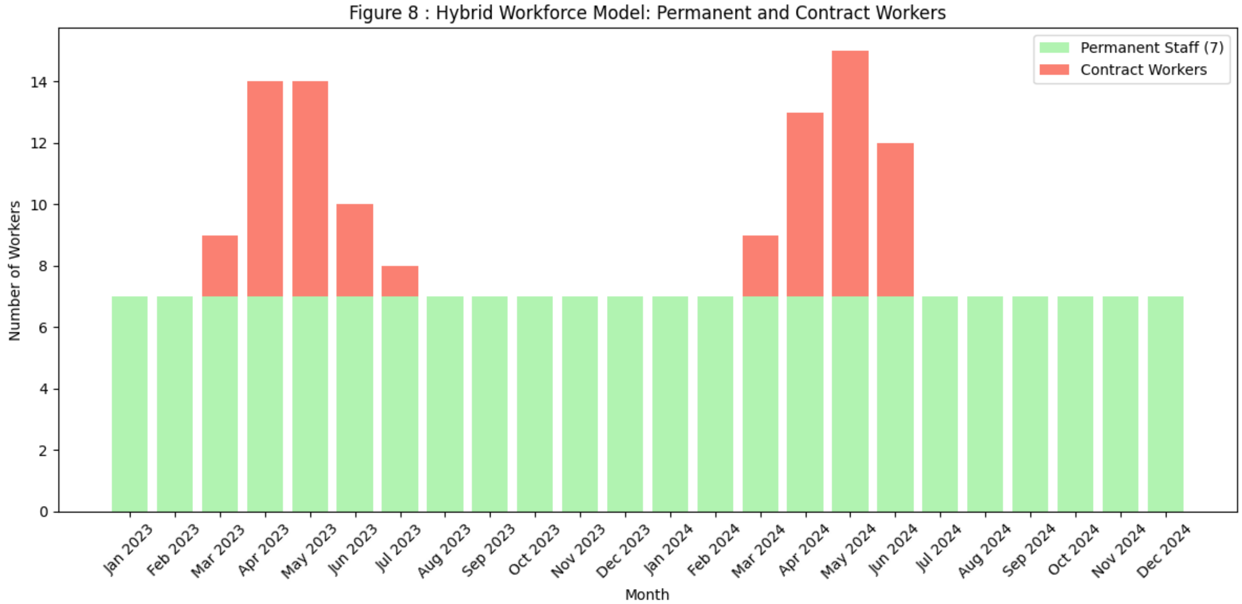


Figure 8 represents a stacked bar chart that shows a constant **permanent staff of 7** supplemented by additional **contract hires** where needed. The total (permanent + contract) represents the hybrid workforce model required to meet monthly demand.

This hybrid approach offers several advantages:

Cost Efficiency: Permanent staff salaries are fixed, while contract hires can be engaged on an as-needed basis, reducing overall labor costs during off-peak periods.

Flexibility: The organization remains agile, scaling contract labor up or down in response to seasonal fluctuations in service demand.

Service Quality: By ensuring the permanent team consistently meets baseline service levels, customer satisfaction is maintained. Contract hires supplement the workforce only when the workload justifies it, helping to prevent both overstaffing and understaffing.

By establishing a stable core of **7 permanent workers** and supplementing with contract hires during periods of elevated demand, the repair center can:

Enhance cost-effectiveness by avoiding the financial burden of overstaffing during off-peak months.

Maintain service quality consistently through the presence of a dedicated permanent team.

Adapt dynamically to fluctuations in service requests, ensuring sufficient staffing during busy periods without committing to an unnecessarily high permanent headcount.

“While the initial LP model aimed to optimize workforce cost through direct minimization, a hybrid staffing strategy combining permanent hires with contract-based flexibility emerges as a more practical and cost-effective solution. This approach supports both operational efficiency and customer satisfaction by matching staffing levels with real world demand patterns.”

4. Interpretation of the Results and Recommendation

The detailed analyses implemented for Aone Cooling Center uncovered a series of meaningful key insights that address both the procurement cost inefficiencies as well as seasonal workforce imbalances. With a data-driven approach using spreadsheet analysis, the integrated use of descriptive statistics, time series forecasting (using ARIMA model), and optimization algorithms (Linear Programming model) has transformed our understanding and made a clearer picture of how demand and pricing vary throughout the year, and how staffing interact across different times of the year. The following interpretations and recommendations are drawn directly from the results to inform smarter operational decisions moving forward.

Procurement and Inventory Optimization

Seasonal Demand Patterns

The first key insight arose from analyzing the collective quarterly demand of spare parts over 2023 and 2024. It was observed that April–July consistently marked the peak season, indicating a recurring surge in service requests. Figure 1 represents bar chart comparing combined spare parts demand across the three segments (Jan–Mar, Apr–Jul, Aug–Dec) reveals distinct seasonal peaks. **April–July consistently records the highest demand, with 186 units in 2023 and 183 in 2024**.

Although the overall demand in 2024 is slightly lower than in 2023 during the other periods, the pattern remained intact, emphasizing that certain times of the year place significantly higher strain on inventory. This insight is crucial because it helps identify when the business must ramp up inventory levels to ensure timely service.

Recommendation –
To avoid shortages during peak seasons, procurement cycles should be scheduled actively, most preferably in the months before the high-demand phase. A lean restocking model can be followed in off-peak quarters to reduce holding costs. Procure critical components such as refrigerants and valves by the end of the previous year, ideally during October to March when demand and prices are lower.

Variation in Unit Prices and Strategic Timing of Purchases

Complementing the demand analysis was the examination of unit price trends. The unit price comparison across quarters that is shown in Figures 2 and 2.1 indicated that prices often rise in April–July, matching the demand surge. Most notably R-22, for example, showed a ₹100 increase during this period compared to its cost in other seasons. Such seasonal increment in prices, if ignored, can severely impact budget planning.
The convergence of these factors suggests that unplanned purchases during busy periods result in both delayed service and elevated spending.

Recommendation –
It is advisable to bulk purchase high cost and high demand components during the low phase of January to March period when prices are relatively stable. For instance, ordering R-22 in Q1 rather than Q2 would save approximately ₹100 per unit.
Additionally, maintaining a buffer stock of stable-demand items (e.g., copper rods, drier filters) can help absorb price changes without overstocking.
Lock in vendor prices ahead of seasonal hikes through annual contracts or bulk purchasing deals during low-demand periods.

Forecasting Spare Part Demand Using ARIMA

To better align procurement with future requirements, a Time Series Forecasting model ARIMA was implemented using historical demand data for key components. A specific case study on R-22 was conducted to demonstrate this application, as shown in **Figure 3**. The forecast for Q1 and Q2 2025 suggested a modest decrease in demand after a previous seasonal peak.

The stable yet slightly decreasing forecast for Q1–Q2 2025 indicates that procuring spare parts in advance (preferably during the lower-priced season, i.e., before April) is essential to prevent cost overruns and avoid supply shortfalls.

Recommendations –

Advance Purchasing:
Schedule procurement of high-demand and high-cost items (such as R-22, Copper rod, and Hydrocarbon) during the low-cost months (January–March) to capitalize on the lower unit prices and avoid the mid-year surge.

Tailored Inventory Management:
Develop individualized procurement strategies for each spare part. Items with forecasted lesser or near zero demand (e.g., 45 Rymning and Welding hand) should not be prioritized, while high demand items are ordered in bulk.

Dynamic Forecasting Integration:
Continuously updating the ARIMA model with new data to refine forecasts, ensuring that purchasing decisions remain aligned with developing market conditions and seasonal trends.

Workforce Optimization and Hybrid Staffing Strategy

Visualizing Service Demand and Staffing Capacity

The analysis of monthly service requests over two years exposed a mismatch between actual service workload and the fixed staffing of 8 workers per month. **Figure 4** shows monthly service volumes, and **Figure 5** shows details about hired workers vs required workers.

During months like April and May, service requests peaked significantly, requiring up to 14 workers. Conversely, months such as December had very low service load, where 4 workers would have sufficed. However, the static staffing model led to overpayment during low seasons and staff shortages during peak periods, adversely affecting operational efficiency.

To better understand, **Figure 6** shows the information about workers in excess or in shortage each month. Negative values indicate shortages, while positive values denote excess staff. This visual representation reflects operational inefficiencies where either service quality is compromised or salaries are spent on sitting manpower.

Recommendation –
Align labor costs with actual demand to reduce idle sitting wages and minimize cost overruns.

Ensure that staffing levels are enough to avoid service delays, thereby enhancing customer satisfaction and maintaining the quality of service.
Track monthly worker demand and develop a responsive hiring mechanism. Short-term hiring (or contract work) during peak months will address shortfalls without affecting the permanent staff structure.

Optimization Using LP and the Need for Practical Adjustments

An LP-based optimization model shown in Figure 7 was initially applied to determine the cost-effective number of workers per month. However, while mathematically optimal, its practical application was limited as one cannot do such frequent hiring and release of staff in such an early manner feasibly.

Thus, the LP model served better as a benchmark to determine the median required workforce level, which turned out to be 7 workers. This observation laid the foundation for a new strategy.

Recommendation –
Implementing a Hybrid Staffing Strategy

Implementing a Hybrid Staffing Strategy

To maintain a balance between consistency and flexibility, a **hybrid staffing model** is proposed and illustrated in **Figure 8**. Under this model:
A fixed team of **7 permanent staff** ensures coverage during regular months.
During peak periods, **contractual workers** are brought in temporarily to meet surging demand.

The LP model's output shown in Figure 7 and the stacked bar chart for the hybrid workforce shown in Figure 8 demonstrate that the optimal solution is to maintain a core permanent staff of 7 workers and supplement with contract hires during high-demand months. This hybrid approach delivers the required flexibility while also preventing the hazard of overpaying in non-peak periods.

Recommendations:

Adopting a Hybrid Staffing Model:
Maintain a permanent core of 7 skilled technicians to reliably handle routine workload. For months where the calculated demand exceeds this baseline, supplement with contract workers. For instance, if 14 workers are needed, hire 7 additional contract workers.

Dynamic Resource Allocation:
Use the historical demand analysis as well as the LP model forecasts to plan contract hires ahead of requirement. This data-driven approach minimizes wasteful expenditure during periods of low demand while ensuring sufficient capacity during peaks.

Regular Reassessment:
Periodically update staffing requirements and forecasts by incorporating the most recent service request data. This will help fine-tune the balance between permanent and contract staffing as the business environment evolves.

Strategic Outlook and Future Directions

The integration of demand, cost analysis, and staffing optimization in this project marks a significant shift from intuition-based management to analytical decision-making at Aone Cooling Center. The findings suggest that proactive planning can yield substantial cost savings while maintaining and improving service standards. The integrated analysis underscores the importance of a two strategy to enhance operational efficiency:

For Procurement:
Data driven forecasts and seasonal cost analyses enables the repair center to schedule purchases before the start of the peak pricing period, maintain optimal inventory levels, and avoid both under stocking and over stocking situations.

For Workforce Management:
A hybrid staffing strategy that combines a stable permanent team of 7 highly skilled workers with flexible contract-based hires provides a scalable and cost-effective solution to handle seasonal up and downs in service demand.

By incorporating these recommendations, Aone Cooling Center can ensure sustainable financial performance, improved service reliability, and a robust competitive edge in managing both procurement and worker managing costs.